Lesson Notes

MODULE 2 | LESSON 3

OPTIMAL PORTFOLIOS AND CURRENCY MANAGEMENT

Reading Time	120 minutes
Prior Knowledge	Portfolio optimization, Financial instruments like forward contracts, options (vanilla and exotic), etc.
Keywords	Currency risk, Calculating portfolio return in domestic currency, Risk of portfolio in domestic currency, Bid/ask spread for currencies, Interest rate parity, Purchasing power parity, Passive hedging, Discretionary hedging, Active hedging

In the previous lesson we look at how to optimize portfolios based on more than just mean and variance and introduced skewness, kurtosis, and liquidity. In this lesson we add an additional element of complications: foreign assets and currency risk.

1. Generalities on Working with Currencies

A well-diversified portfolio is likely to consist of domestic assets (denominated in the investor's home currency) and foreign assets (denominated in a currency different from the home currency). This creates additional complications for both risk and returns. At a minimum, leaving aside specific country risks, when an investment is priced in a currency different from the domestic one, there are two sources of risk:

- the return on the foreign asset denominated in the foreign currency;
- the return on the foreign currency, which derives from the change in the exchange rate between the domestic and the foreign currency.

Let r_D and r_F represent the return from investing in an asset denominated in domestic and foreign currency, respectively, and let r_{FX} be the rate of change in the foreign currency vs. the domestic currency. Then, we have

$$r_D = (1 + r_F)(1 + r_{FX}) - 1 \approx r_F + r_{FX}.$$

In some instances, r_{FX} could be known, but often it is not. This creates a need to hedge the global (domestic+foreign) portfolio. If we decided that we want to hedge the currency exposure, we would need to sell short the ending amount of foreign currency units, which is not known in the case of risky assets and makes managing currency risk complicated.

Example (Calculate returns in domestic currency): Consider an Italian investor (home currency is EUR) who invests in a portfolio of natural resources that trade in Canadian dollars (foreign currency is CAD). Over a one-year period, the value of the investor's portfolio has increased by 10% and the CAD-EUR exchange rate went from 1.30 to 1.20. What is the investor's return in her domestic currency?

Note that the exchange rate went from 1.3 CAD for 1 EUR to 1.2 CAD for 1 EUR, i.e., the euro has depreciated vs. the Canadian dollar, which is good news for the investor as she gets paid in the foreign currency. More specifically, we went from 1.3 CAD for 1 EUR to 1.2 CAD for 1 EUR or .769 EUR for 1 CAD to 0.833 EUR for 1 CAD, i.e., the foreign currency had a return of 8.32% vs. the domestic one. Then, using the formula above, we find that

$$r_{EUR} = (1 + 10\%)(1 + 8.32\%) - 1 = 19.2\%.$$

Using the approximate formula, we would find

$$r_{EUR} = 10\% + 8.32\% = 18.32\%$$

so the formula is not always accurate unless the returns on both the assets and the currencies are small.

In the case just depicted, the exchange rate moves in the investor's favor, but if the exchange rate had gone from 1.30 to 1.40 instead, then the EUR would have appreciated vs. the Canadian dollars (bad news for the investor who is paid in Canadian dollars).

In this case we go from receiving 0.769 EUR for each CAD to receiving 0.714 EUR for each CAD, a return on the foreign currency of -7.12%. Hence:

$$r_{EUR} = (1 + 10\%)(1 - 7.12\%) - 1 = 2.17\%,$$

a dramatic difference when compared to the previous case.

While pure quants are not usually tasked with managing portfolios and currency risk, they may have to support a portfolio manager, and nothing stops them from potentially becoming portfolio managers if they so desire. Therefore, it is important to understand currency risk. For this reason, we will devote the remainder of the lesson to understanding currency risk and how to manage it.

2. Basics of Exchange Rates

To deal with exchange rate movements, one must understand what causes exchange rates to move in the first place. We will not devote a lecture to International Economics, but we will summarize the key issues so that a student can understand and be competent in the essential elements of exchange rates.

We start with some terminology. The base currency is the denominator of the exchange rate, and it is priced in terms of the currency in the numerator.

So, for instance, if we quote CAD/USD = 1.25, we mean that 1 USD is worth 1.25 CAD.

We will assume that the base currency is the domestic currency for what we do in this lesson. This has the merit of establishing a clear way to quote currencies vs. other currencies. In general, currencies are quoted as bid/ask spreads. For example:

	bid	ask
Spot CAD/USD	1.2570	1.2580

How do we read this table for practical purposes? Say that a U.S. investor has USD $1,000,000 to convert to CAD. Then, the investor is buying CAD and selling USD. Therefore, the bid price is the relevant one, and the investor will get CAD 1,257,000. Now, assume that the same investor receives CAD 1,500,000 from an investment in Canada and needs to convert the amount to USD. In this case, the investor is selling CAD and buying USD, and the relevant price is the ask price. Then, the investor will get USD $1,500,000/1.2580= 1,192,368.84. Often, the easiest way to remember how it works is to think that one always gets the worst of the two prices for what they need to buy/sell!

Currency quotes can also be defined as **direct** or **indirect**. An indirect quote is when the foreign exchange rate is reported in terms of units of foreign currency per unit of domestic currency. This is what we have done above. A direct quote is when currency exchange is expressed in terms of units of domestic currency per unit of foreign currency. The relationship between the two is very simple:

$$\text{direct quote} = \frac{1}{\text{indirect quote}} \quad \text{and} \quad \text{indirect quote} = \frac{1}{\text{direct quote}}$$

Make sure that you are aware of which one you are given while reading quotes or scholarly papers.

Now that we know how to read currency quotes, we ask ourselves the following question: What moves the exchange rates between two currencies? We will consider the two main models.

2.1 Interest Rate Parity Model

This model proposes a relationship between the interest rates of two given currencies and the spot and forward exchange rates between the currencies. As such, it can be used to predict how exchange rates between two currencies change when the risk-free interest rates of the two currencies are known. This theory is based on the assumption that no arbitrages should exist, not for long at least, and that is also why it is considered a model for the short/medium period. The forward exchange rate is the exchange rate at which a party is willing to enter into a contract to receive or deliver a currency at some future date. Essentially, this is what matters as one makes certain investments to receive future benefits or as a manufacturer may purchase items to be delivered and paid for in the future. Let S be the spot rate (the exchange rate right now) and let F be the forward rate (the exchange rate in the future). Furthermore, let r_d be the interest free rate of return on deposits in the domestic currency and r_f the interest free rate of return on deposits in the foreign currency. Then, if n stands for the number of years, the formula that links F and S is the following:

$$F = S \cdot \left(\frac{1 + r_f}{1 + r_d} \right)^n.$$

Example: Let's assume that the current exchange rate between Japanese Yen (JPY) and USD is 115.70. Also, assume that the interest rate in Japan is 4% and the interest rate in the U.S. is 2%. What do we expect will happen to the forward exchange rate between the two currencies in a one-year time? Well, $S = 115.7, r_f = 4\%, r_d = 2\%, n = 1$ and using the formula given earlier:

$$F = 115.7 \times \left(\frac{1+4\%}{1+2\%}\right)^1 = 117.97,$$

i.e., the Yen is depreciating. This is natural because, otherwise, there would be an arbitrage opportunity. For instance, assume that the forward exchange rate were 110. A U.S. investor could walk to a bank in the U.S. and borrow USD $1,000,000 for 2% interest rate. Then, the investor can convert the money to JPY at the spot rate. Therefore, the investor receives YEN 1,000,000 x 115.70 = 115,700,000, which can be invested in the foreign country at 4%. After one year, the U.S. investor receives JPY 115,700,000 x 1.04 = 120,328,000. At this point, the investor converts back to USD using the forward rate, i.e., the investor gets USD 120,328,000/110 = 1,093,890.91. Now the investor pays the interest rate on the USD $1,000,000 borrowed from the bank, i.e., USD $20,000 and has netted a profit of USD $73,890.91. Hence, the only way to avoid arbitrage is if JPY depreciates.

2.2 Purchasing Power Parity Model (PPP)

In this case, the exchange rate of any two currencies will remain equal to the ratio of their respective purchasing powers. As usual, the purchasing power of a currency is measured as the amount of the currency needed to buy a selected product or basket of goods commonly available in different countries, and it relates to inflation. So, if we define by i_f and i_d the inflation rate in the foreign and domestic country, respectively, we have the following formula:

$$F = S \cdot \left(\frac{1+i_f}{1+i_d}\right)^n.$$

Example: Assume that the exchange rate GBP/USD = 0.75. Also, assume that the price of a standardized basket of goods is $20,000 in the U.S. and GBP 15,000 in the UK. Furthermore, we believe that in one year, the same basket will cost $22,000 in the US and GBP 16,000 in the UK. Hence, $i_d = i_US = 22,000/20,000 - 1 = 10$ and $i_f = i_UK = 16,000/15,000 - 1 = 6.67$ Then,

$$F = 0.75 \cdot \left(\frac{1+6.67\%}{1+10\%}\right)^1 = 0.7273,$$

i.e., GBP will appreciate against USD.

3. Calculating Portfolio Return for Investments in Foreign Assets

If we assume that an investor owns assets across multiple markets with different currencies, the first task is to understand how to compute the return on such a portfolio. As the return is a weighted average of returns, we need to how compute weights and returns. It seems only logical that:

- the weights are computed as the percentage of the portfolio invested in each foreign currency denominated asset expressed in terms of domestic currency.
- the returns are also expressed in terms of the domestic currency.

Thus,

$$r_p^{(d)} = \sum_{j=1}^{m} w_j r_j^{(d)},$$

where each index value of j refers to a different currency. There is no problem in aggregating all assets in the same currency as return is a linear function.

Example: Suppose that we have an investor based in Canada and that this investor has GBP 1,000,000 invested in UK assets, EUR 2,000,000 invested in Germany, and USD $3,000,000 invested in US assets.

We are also given that $r_{US} = 10\%, r_{EUR} = 8\%,$ and $r_{GBP} = 7\%.$ In addition we are also told that

$$\textbf{beginning CAD/GBP} = 1.50 \quad \textbf{ending CAD/GBP} = 1.55;$$

$$\textbf{beginning USD/CAD} = 0.75 \quad \textbf{ending USD/CAD} = 0.78;$$

$$\textbf{beginning EUR/CAD} = 0.70 \quad \textbf{ending EUR/CAD} = 0.68.$$

Task 1. Translate all assets in domestic (CAD) currency at the beginning of the period:

$$A^{GBP} = 1,000,000 \cdot (1.50) = CAD1,500,000;$$

$$A^{EUR} = 2,000,000 \cdot (1/0.70) = CAD2,857,143;$$

$$A^{USD} = 3,000,000 \cdot (1/0.75) = CAD4,000,000.$$

Task 2. The total amount invested in the international portfolio is $A^{CAD} = 8,357,143.$ This allows us to compute the weights:

$$w_{GBP} = 1,500,000/8,357,143 = 0.179, \quad w_{EUR} = 2857,143/8,357,143 = 0.342,$$

$$w_{USD} = 4,000,000/8,357,143 = 0.479.$$

Task 3. Compute the returns for each class of assets in domestic currency (CAD):

$$r_{GBP}^{CAD} = (1+7\%)(1+(1.55/1.50-1)) - 1 = 1.07 \cdot 1.0333 - 1 = 10.56\%;$$

$$r_{EUR}^{CAD} = (1+8\%)(1+((1/0.68)/(1/0.70)-1)) - 1 = 1.08 \cdot 1.0294 - 1 = 11.18\%$$

$$r_{USD}^{CAD} = (1+10\%)(1+((1/0.78)/(1/0.75)-1)) - 1 = 1.10 \cdot 0.9615 - 1 = 5.77\%$$

Note that the CAD appreciated against the USD and depreciated against both GBP and EUR.

Task 4. Compute the return in domestic currency:

$$r_P^{CAD} = 0.179 * 10.56\% + 0.342 * 11.18\% + 0.479 * 5.77\% = 8.48\%.$$

4. Portfolio Risk for Investments in Foreign Assets

Investments involving foreign denominated assets have two sources of risk: the change in price of the investments (in the foreign currency) and the fluctuations in the exchange rate between the

foreign currencies and the domestic one.

The first task awaiting us is to define the volatility of a foreign denominated asset in the international portfolio. For each foreign denominated asset j, using the approximation

$$r_j^{(d)} \approx r_j^{(f)} + r_j^{(fx)}$$

we derive the following formula for the variance:

$$\sigma^2(r_j^{(d)}) \approx \sigma^2(r_j^{(f)}) + \sigma^2(r_j^{(fx)}) + 2\sigma(r_j^{(f)})\sigma(r_j^{(fx)})\rho(r_j^{(f)}, r_j^{(fx)})$$

where $\rho(r_j^{(f)}, r_j^{(fx)})$ is the correlation for the return of the asset j in the foreign currency and the fluctuations of the currency of asset j itself.

Similarly, $\sigma(r_j^{(fx)})$ is used to denote the volatility of the currency for asset j. The notation is arguably cumbersome, but it shows that the variance for the return of asset j in the domestic currency is higher than the return of asset j in its foreign currency.

However, if the correlation for the return of the asset j in the foreign currency and the fluctuations of the currency of asset j is negative, the volatility of the returns in the domestic currency is dampened by it.

Now, if we use the formula for $\sigma^2(r_j^{(d)})$ just introduced, we can define the variance for the portfolio as

$$\begin{aligned}\sigma_P^{2,(d)} &= \sigma^2\left(\sum_{j=1}^{m} w_j r_j^{(d)}\right) \\ &= \sum_{j=1}^{m} w_j^2 \sigma^2(r_j^{(d)}) + 2\sum_{j=1}^{m}\sum_{h>j}^{m} w_j w_h \sigma(r_j^{(d)})\sigma(r_h^{(d)})\rho(r_j^{(d)}, r_h^{(d)}).\end{aligned}$$

To complete this section, we need one last effort. One notices the term $\rho(r_j^{(d)}, r_h^{(d)})$, and we need to write it in terms of $r_j^{(f)}$ and $r_h^{(f)}$ and the exchange rates. Recalling the definition of correlation, the denominator is not a problem as we already have $\sigma^2(r_j^{(d)})$ and $\sigma^2(r_h^{(d)})$, and can concentrate on the covariance term in the numerator.

Thus, using the bilinearity property of the covariance operator, we find

$$\sigma(r_j^{(d)}, r_h^{(d)}) = \sigma(r_j^{(f)}, r_h^{(f)}) + \sigma(r_j^{(f)}, r_h^{(fx)}) + \sigma(r_h^{(f)}, r_j^{(fx)}) + \sigma(r_j^{(fx)}, r_h^{(fx)}).$$

Therefore,

$$\begin{aligned}\rho(r_j^{(d)}, r_h^{(d)}) &= \frac{\sigma(r_j^{(d)}, r_h^{(d)})}{\sigma(r_j^{(d)})\sigma(r_h^{(d)})} \\ &= \frac{\sigma(r_j^{(f)}, r_h^{(f)}) + \sigma(r_j^{(f)}, r_h^{(fx)}) + \sigma(r_h^{(f)}, r_j^{(fx)}) + \sigma(r_j^{(fx)}, r_h^{(fx)})}{\sigma(r_j^{(d)})\sigma(r_h^{(d)})}\end{aligned}$$

with $\sigma(r_j^{(d)})$ and $\sigma(r_h^{(d)})$ defined above.

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5. Optimal Portfolio with Multiple Currencies

Not unlike what we saw in previous lessons, optimization in the case of a portfolio that involves multiple currencies requires that one selects weights that identify a portfolio on the efficient frontier for the best trade-off between mean and variance. To achieve this, we need to estimate expected returns and volatilities of each of the foreign currency denominated assets. This adds a lot of parameters to our model as we would need the returns of each asset in the foreign currency, $r_j^{(f)}$ for $j = 1, \ldots, m$, the returns on the foreign currency of each of the assets, $r_{j,fx}$, volatility of foreign denominated returns, $\sigma(r_j^{(f)})$, volatility on the exchange rates $\sigma(r_{j,fx})$, as well as many correlations.

However, if we could do that, then the problem would be no different than what we have seen already. In its simplest version:

$$\begin{aligned} &\max_w w^T r^{(d)} - \tfrac{1}{2} w^T \Sigma^{(d)} w \\ &\text{subject to} \quad w^T \iota = 1 \end{aligned}$$

where $r^{(d)}$ and $\Sigma^{(d)}$ represent the parameters expressed in domestic currency and that were defined in the previous section.

Estimating $r^{(d)}$ is not a big deal computationally (although estimates could be unreliable); estimating $\Sigma^{(d)}$ is cumbersome.

An exact and computable solution to the general quadratic mean-variance problem that incorporates exchange rate and return volatility can be obtained following an approach by Rustem (1995). We present the relevant steps:

Step 1. Let r^i represent the return vector on n^i investments in currency i and assume that

$$r^i = E[r^i] + \rho^i, \;\; \rho_i \sim N(0, \Lambda^i_{\rho\rho}), i = 1, 2, \ldots, I$$

where $\Lambda^i_{\rho\rho}$ is the $n^i \times n^i$ variance covariance matrix for returns in currency i.

Step 2. Let e^i be the exchange rate associated with currency i, i.e.

$$e^i = E[e^i] = \epsilon^i \;\; \epsilon_i \sim N(0, \Lambda^i_{\epsilon\epsilon}), i = 1, 2, \ldots, I$$

where $\Lambda^i_{\epsilon\epsilon}$ is the $n^i \times n^i$ variance covariance matrix for returns in currency i.
The exchange rate corresponding to the base currency would be equal to 1 and its variance-covariance matrix is the null matrix.

Step 3. Let r and ρ denote the $\sum_{i=1}^{I} n^i \times 1$ vectors of returns on all assets and the corresponding uncertainty, that is

$$r = E[r] + \rho.$$

Step 4. Let e by the $I \times 1$ vector of all exchange rates, and ϵ their uncertainty, i.e.,

$$e = E[e] + \epsilon.$$

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Step 5. Now let

$$\beta = \begin{pmatrix} \rho \\ \epsilon \end{pmatrix} \sim N(0, \Lambda)$$

with

$$\Lambda = \begin{bmatrix} \Lambda_{\rho\rho} & \Lambda_{\rho\epsilon} \\ \Lambda_{\epsilon\rho} & \Lambda_{\epsilon\epsilon} \end{bmatrix}.$$

An estimate for $\Lambda_{\rho\rho}$ can be obtained from historical returns data; for $\Lambda_{\epsilon\epsilon}$, we can use the error between past
forecasts and historical exchange rates. Finally,$\Lambda_{\rho\epsilon}$ can be obtained using historical returns and forecast errors.

Step 6. This is the key step to be able to write the problem in a convenient matrix form. To this purpose, we introduce a matrix q whose dimensions are $\sum_{i=1}^{I} n^i \times I$ **:** defined as follows:

$$q_{ij} = \begin{cases} 1 & \textbf{if investment i is in currency j} \\ 0 & \textbf{otherwise.} \end{cases}$$

Note that if we have one investment for each currency, then $Q = I$, the unit matrix.

Example: Suppose we have stocks in three different currencies, A, B, and C. Two stocks are denominated in currency A, 1 in currency B, and 2 in currency C. What is the q matrix in this case? In this case,$n^1 = 2, n^2 = 1, n^3 = 2.$ Therefore,q is a5×3 matrix that looks like this:

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 7. Introduce one more matrix:

$$P = \mathrm{diag}(Qe)$$

which is of dimension $\sum_{i=1}^{I} n^i \times \sum_{i=1}^{I} n^i$ and whose diagonal i-th diagonal element is given by the i-th element of $Qe.$

Example: Using the setup of the previous example, if $e^A, e^B,$ and e^C denote the exchange rates of the foreign currencies vs. the domestic one, we have

$$Qe = \begin{pmatrix} e^A \\ e^A \\ e^B \\ e^C \\ e^c \end{pmatrix} \quad \text{and} \quad P = \begin{pmatrix} e^A & 0 & 0 & 0 & 0 \\ 0 & e^A & 0 & 0 & 0 \\ 0 & 0 & e^B & 0 & 0 \\ 0 & 0 & 0 & e^C & 0 \\ 0 & 0 & 0 & 0 & e^C \end{pmatrix}$$

Step 8. Equipped with all this, the return in the base currency can be written as

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$$R = Pr$$

which is a $\sum_{i=1}^{I} n^i \times 1$ vector.

Step 9. Let w be the $\sum_{i=1}^{I} n^i \times 1$ vector of portfolio weights for the multi-currency assets. Then, the return for the portfolio can be written as

$$w^T R = r^T(\text{diag}(w)P)\iota.$$

Step 10. Now from a theorem proved (this proof is not hard) by Rustem, we can write:

$$E[w^T R] = (E[r])^T(\text{diag}(w)PE[e]) + \frac{1}{2}\text{tr}(\Lambda\Omega(w))$$

where $\Omega(w)$ is the following matrix

$$\Omega(w) = \begin{bmatrix} 0 & \text{diag}(w)Q \\ Q^T\text{diag}(w) & 0 \end{bmatrix}.$$

Step 11. Now we get the variance for the returns in the domestic currency using a second theorem proved by Rustem (this is harder):

$$\text{Var}(w^T R) = w^T M^T \Lambda M w + \frac{1}{2}\text{tr}(\Lambda\Omega(w))^2$$

with

$$M = \begin{bmatrix} \text{diag}(QE[e]) \\ Q^T\text{diag}(E[r]) \end{bmatrix}$$

an $(\sum_{i=1}^{I} n^i + I) \times \sum_{i=1}^{I} n^i$ matrix whose first $\sum_{i=1}^{i} n^i$ rows consists of the diagonal matrix whose diagonal elements are taken from the matrix $QE[e]$, and the following I elements are the byproduct of q with a diagonal matrix whose diagonal elements are $E[r]$.

Through the ten steps above, we are now in a position to write the QP to find the optimal portfolio:

$$\begin{gathered} \max_w \ (E[r])^T(\text{diag}(w)PE[e]) + \tfrac{1}{2}\text{tr}(\Lambda\Omega(w)) - \tfrac{1}{2}\left(w^T M^T \Lambda M w + \tfrac{1}{2}\text{tr}(\Lambda\Omega(w))^2\right) \\ \text{subject to} \ \ w^t\iota = 1. \end{gathered}$$

Note that while this is a QP, it is not a standard one. Standard QP look like this (just a reminder as we saw this before):

$$\begin{aligned} \text{minimize} \quad & (1/2)x^T Px + q^T x \\ \text{subject to} \quad & Gx \leq h \\ & Ax = b \end{aligned}$$

6. Strategic Decisions for Portfolio Optimization

In a world where information is complete and freely available, it would be desirable to optimize the portfolio with respect to all these exposures, considering each currency simultaneously.

Unfortunately, in the real world, this is a high task and portfolio managers who are forced to deal with these situations have developed two main approaches:

- fully hedge the portfolio for currency exposure;
- decide which currency they would like to be exposed and hedge the others.

This is possible because the use of derivatives allows us to separate different risk exposures from the other and can manage the price risk (assets selection) separately from exchange-rate risk. Ideally, if there existed a perfect hedge that were free or at least achievable at reasonable costs, currency exposure could be neutralized and, in that case, $\sigma^2(r_j^{(d)}) = \sigma^2(r_j^{(f)})$, our optimization process would be much simpler.

Neither academic research nor empirical analysis produce a strong conclusion about how currency risk is best managed. Among the reasons not to hedge we find:

- hedging is expensive and requires a lot of time;
- in the long run, currencies become a zero-sum game in which if a currency changes against the portfolio, another is likely to change in our favor;
- over time, currencies tend to revert to their (theoretical) fair value and movements away from that value are only temporary.

Practitioners have resorted to three main strategies:

- **Passive hedging:** In this case, portfolio managers limit themselves to matching the currency exposure of the portfolio to that of the benchmark used to evaluate their performance. It does not mean that the portfolio manager does nothing, but rather, their management is limited to periodic re-balancing to keep the exposure in line with that of the benchmark.
- **Discretionary hedging:** This is very similar to passive hedging, except that the portfolio manager is allowed to deviate from the currency's exposure in the benchmark. For example, the manager could be allowed to deviate 5-10% from the exposure in the benchmark.
- **Active management:** This is similar to discretionary hedging, but the manager is allowed more discretion and is expected to generate incremental returns (hopefully positive!)

The essence of the three approaches could be summarized as follows:

- passive hedging = eliminate currency risk relative to benchmark;
- discretionary hedging = reduce currency risk and pursue modest incremental returns;
- active management = generate additional alpha; reducing risk is secondary or not required.

It is not unusual for active management to be outsourced to an external party that specializes in currency management (**currency overlay**).

Regarding active management let's ask ourselves: what happens if we were to hedge everything? To this purpose, consider a simple scenario: We own one foreign asset and we buy currency forward contracts. In this case, the variance of the hedged porttfolio (1 foreign asset, (a), and 1 forward currency contract, (f)) can be written as

$$\sigma_p^2 = w_a^2\sigma_a^2 + w_f^2\sigma_f^2 + 2w_a w_f \sigma_a \sigma_f \rho(a, f),$$

where the symbols have the usual meaning and $\rho(a, f)$ represents the correlation between the foreign asset and the forward currency contract.

If we assign a weight of 1 to the asset, then the previous equation becomes

$$\sigma_p^2 = \sigma_a^2 + w_f^2\sigma_f^2 + 2w_f\sigma_a\sigma_f\rho(a, f),$$

which is ideal for calculating the optimal weight for the forward contract in order to minimize the variance of the portfolio. Simple calculus yields

$$w_f = -\frac{\sigma_a}{\sigma_f}\rho(a, f).$$

This tells us two things. First, it tells us that a higher correlation between asset and currency indicates that the exposure to currency increases the risk (less diversification) and requires a higher minimum-variance hedge ratio. In addition, it also tells us that a higher σ_f (the standard deviation of the currency) decreases the hedge.

Example (Hedging 100% of currency exposure is not always optimal): Assume that we have a portfolio consisting of one foreign denominated asset that has a10% standard deviation. Assume also that the foreign currency has a standard deviation of $15\%,$ and the correlation between the asset and the currency is $0.6.$ Then, according to the formula we just derived, the minimum-variance hedge ratio is equal to $-0.10/0.15 \cdot 0.6 = -0.4.$ In this case, the standard deviation of the hedged portfolio would be

$$\sigma_p = \sqrt{0.10^2 + (-0.4)^2 \cdot 0.15^2 + 2 \cdot (-0.40) \cdot 0.10 \cdot 0.15 \cdot 0.6} = 0.080.$$

Hence, the hedge helped to lower the risk of the portfolio. Now, we choose to hedge 100 of the position, i.e.,$w_f = -1.$ Then, using the same formula yields:

$$\sigma_p = \sqrt{0.10^2 + (-1)^2 \cdot 0.15^2 + 2 \cdot (-1) \cdot 0.10 \cdot 0.15 \cdot 0.6} = 0.1204.$$

In other words, a 100 hedge ratio is not the ratio that minimizes portfolio risk, and, in this example, a fully hedged portfolio has a higher volatility than the unhedged one!

Among active hedging strategies, we could list what is often called the **minimum regret hedge ratio**. In this case, investors hedge 50% of their currency exposure without any consideration of volatilities and correlations. The philosophy behind this approach should be clear: Currency exposure can be a friend (if it moves the investor's way) or a foe (it it moves the other way), and therefore. an investor who is prone to suffer from regret should choose between 0% hedge or 100% hedge, but does not know which one is the right course of action. Going for 50% minimizes the investor's *ex post regret*. This approach has been criticized because it does not take into consideration either the probability distribution of possible outcomes or the change in utility when the hedging strategy outperforms or underperforms. In addition, there is a lack of valid empirical studies to test this particular strategy.

7. Hedging Currency Exposure: Implementation

We need to distinguish between two large categories of hedging: linear and non-linear strategies. A linear strategy is one where the return of the hedged portfolio is a linear function of the return on the hedged currencies. Typically, a linear hedge neutralizes both potential losses and potential gains. A non-linear hedging strategy, on the other hand, protects against losses but preserves the potential upsides in the portfolios due to currency exposure and usually refers to using options.
Among the linear strategies, we can list approaches like un-hedged, full-hedged, currency-specific hedging, and more.

There is also a choice among a strategy where the hedge is fixed (**static hedge**) and one in which the hedge changes over time (**dynamic hedge**). In the latter case, we could possibly need to perform heavy re-balancing.

We should look now at how active hedging is implemented with forward contracts. Forward contracts are usually the preferred hedging vehicle since forwards can be customized, forwards are available for almost any currency and in any size, and trading volume is very high.

Say that we have a long exposure to Canadian dollars (CAD) that we would like to hedge. If we expect that the CAD will appreciate, we can reduce our hedge to be hedging less than full exposure (never lose sight that hedging is expensive and not free). On the other hand, if we expect CAD to depreciate, we would increase our hedge. This strategy creates **positive convexity** (increase gains and reduce losses). As far as implementing the hedging, we could use one of the following strategies:

- Buy a put option at the money (portfolio insurance). In this case, we have high initial costs, but no opportunity costs.
- Same strategy but using out-of-the-money puts. This is less expensive, but there is downside up to the strike price of the puts.
- Cheaper still would be to use a collar: For instance, we can buy 35-delta puts on the CAD and sell 35-delta calls on the CAD. This gives us downside protection while removing some upside potential, but it reduces the cost of hedging.
- A put spread where we buy a 35-delta put and sell a 25-delta put. The downside protection does not begin until the strike of the puts we purchased, it is relatively cheap to set up, but if the CAD falls below the strike of the 25-delta put, all the protection is gone.
- A seagull spread is achieved by going long a 35-delta put and a 25-delta put and selling a 35-delta call. This lowers the cost of the hedging strategy even more and provides the same protection as a put spread, but we give up upside potential.
- Knock-in/knock-out options, which provide the same protection/upside of plain vanilla options but come alive only if the CAD falls below a certain threshold or ceases to exist if the CAD surges above a specific threshold (no need for protection anymore).

The table below summarizes a few key facts about hedging currencies.

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ELEMENT	EVENT	ACTION
Currency price	Appreciate	Reduce hedge
	Depreciate	Increase hedge
Volatility	Rises	Long straddle, strangle
	Decreases	Short straddle, strangle
Economy/Market	Stable	Carry trade
	Deteriorates	Stop carry trade

The performance of option strategies can be evaluated via Monte Carlo simulations or back-testing. However, in line with what we have seen in this module so far, we should keep in mind that back testing relies only on one specific outcome (what occurred, i.e., one of possibly infinite outcomes), and Monte Carlo simulations allows us to test our strategy against a large number of possible outcomes.

8. Conclusion

In this lesson, we saw that when a portfolio contains assets denominated in foreign currencies, there are a few complications that one has to deal with. We looked at how to properly read exchange rates, how to calculate their impact on portfolio returns and volatility, and what changes if the exposure to currency is hedged or not hedged. Finally, we considered a few hedging strategies. Understanding these nuances is important to knowing how to work with problems involving foreign assets and/or assets denominated in foreign currencies.

In the next lesson, we will look at factor models and what their implications are, both in terms of reducing the parameters to estimate and as a tool in and of themselves.

References

- Rustem, B. "Computing Optimal Multi-Currency Mean-Variance Portfolios." *Journal of Economics Dynamics and Control*, vol. 19, 1995, pp. _901–908, _ https://doi.org/10.1016/0165-1889(94)00812-V